

Ansh Singh

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SUMMARY

Results-driven Data Scientist and ML Engineer specializing in end-to-end AI lifecycles. Expert in architecting computer vision (CNNs) and NLP solutions, backed by strong MLOps proficiency (MLflow, Docker, AWS). Proven ability to transition complex models from research into scalable, production-ready REST APIs.

EDUCATION

Vellore Institute of Technology, Chennai (VIT Chennai)
Bachelor of Technology in Computer Science and Engineering

2023 – 2027
Chennai, TN

EXPERIENCE

Centre for Smart Grid Technologies Research Intern | *CSGT, VIT Chennai* (Remote) **May 2026 – July 2026**

- Developed a multi-class brain tumour classification machine learning pipeline for MRI scan analysis using **ResNet50**, **DenseNet121**, and **EfficientNetB5** architectures, achieving **95.00%** accuracy.
- Conducted a 12-model ablation study comparing Squeeze-and-Excitation (SE), CBAM, and Hybrid attention mechanisms to optimize diagnostic performance.
- Integrated Explainable **AI (XAI)** using **Grad-CAM** visual saliency maps to audit model decision pathways, identifying **20+** edge-case failures and boosting clinical trust by **35%**.
- Co-authored a research paper** presenting the framework, ablation study, and XAI evaluation, submitted to a **Q1/Q2 international journal** for peer review.

TECHNICAL SKILLS

- Languages:** Python, SQL, C++, C, Java, JavaScript (ES6+), TypeScript, Bash
- Data Science & Analytics:** NumPy, Pandas, Matplotlib, Seaborn, Statistical Modeling, EDA, Data Preprocessing
- Machine Learning & AI:** Scikit-learn, Deep Learning, NLP, Computer Vision, TensorFlow, PyTorch, LightGBM
- MLOps & Cloud:** MLflow, DVC, Docker, AWS (EC2, S3, CodeDeploy), CI/CD, Model Registry
- Backend & APIs:** Node.js, FastAPI, RESTful APIs
- Databases & Tools:** MongoDB, MySQL, PostgreSQL, Redis, Git/GitHub, Postman, Linux

PROJECTS

PlantGuard-AI | *Python, TensorFlow, CNN, ResNet-50, Matplotlib* GitHub Repository

- Designed a high-accuracy deep learning system to detect crop diseases across plant species using custom CNNs and ResNet-50 transfer learning, achieving **96% validation accuracy**.
- Built species-specific pipelines with targeted data augmentation strategies (rotation, flipping, scaling) to enhance model generalizability and reduce class bias.
- Delivered an **8% accuracy improvement** over baseline CNN through architecture refinement and augmentation tuning.

Brain Tumor MRI Classification with Attention Mechanisms & XAI | *TensorFlow, Grad-CAM* GitHub Repository

- Classified brain tumors from MRI scans across **12 model variants**, achieving peak test accuracy of **98.00%** (DenseNet121 + SE attention).
- Benchmarked SE, CBAM, and hybrid attention mechanisms across 3 backbones and validated model explainability using **Grad-CAM** with ROAD metric scoring (mean: **0.4205**).
- Engineered a reproducible dual-environment pipeline (**local GPU + Google Colab**) with a custom checkpoint callback resolving a **TensorFlow 2.10** serialization bug.

Comment Sentiment Analysis | *NLP & MLOps / LightGBM, MLflow, DVC, AWS, Docker* GitHub Repository

- Designed an NLP pipeline using **NLTK** and TF-IDF vectorization (10k features) to train an optimized **LightGBM** classifier for multiclass sentiment analysis.
- Established an **MLOps workflow** with **MLflow** hyperparameter tracking and **DVC** data version control.
- Containerized inference with **Docker** and deployed a Flask API to **AWS (EC2, CodeDeploy)** for real-time batch predictions.

COURSES & CERTIFICATIONS

- [Google Cybersecurity] — Google [DevOps Fundamentals] — IBM [DevOps, Agile & Design Thinking] — IBM
- [Structured Query Language] — CU Boulder [JS Basic] — GeeksforGeeks / HackerRank

ACHIEVEMENTS

- Competitive Programming:** Solved **500+ problems** (LeetCode, GFG, Codeforces); **Max Rating 1700+ (Top 10% Globally)**.
- Contest Rank:** Secured **Global Rank 944** in LeetCode Biweekly Contest 162 and 848 in Biweekly Contest 159.
- Community:** Finalist at **FlutterSprint Hackathon 2K26** & **Google Developer Club**; active contributor to open-source web accessibility tools.
- Soft Skills:** Strong communicator with cross-functional team collaboration experience.